

To What Extent Has the Adoption of AI and Automation Reduced Demand for Mid-Skill Labour in Service Industries?

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ABSTRACT

Historically, the impact of artificial intelligence (AI) and automation tools on mid-skill workers – workers situated middle skilled on the wage distribution and who typically work on routine, codifiable activities – has attracted both scholarly and policy interest, with the potential scale of job losses among these U.S. workers more pronounced in service sectors compared to manufacturing. Based on secondary data from the World Economic Forum, the McKinsey Global Institute, the International Labour Organization, OECD, Goldman Sachs Research and leading academic economists such as Daron Acemoglu, David Autor, and their associates, this paper looks at how much AI and automation have taken the place of mid-skill jobs in services. The evidence shows an enduring trend of job polarization: the ratio of middle-wage to high-skill and low-skill jobs has steadily fallen since the 1980s and has been driven by the routine-biased technological change (RBTC). The most stark example of what the service sector showcases today are bank tellers, customer service representatives, data-entry clerks and administrative support workers whose prospects for jobs over the next 10 years are predicted to decline by 5–15 per cent – and corporate news has plenty of examples of massive upheavals like the one at Klarna, where the company has in one year slashed its administrative workforce by 24 per cent, installing new algorithms instead. Meanwhile, at the same time, authoritative players like the ILO and OECD counsel against the simple substitution story. Few jobs are fully automated, and there is a greater number of jobs that are

fundamentally created jobs with human perform that has a causal role extending to less closely automated jobs. Whole automation is less common than partial, there is a relatively smaller number of jobs that are fundamentally created jobs with human work which has a causal role extending to less closely automated jobs, which is the concept of augmentation. In addition, the WEF's 2025 Future of Jobs reports a net global gain of 78 million jobs somewhere around 2030 together with the displacement of 92 million jobs. In the end, the effect is substantial but uneven – STEM jobs' surplus of workers is real, empirically described, and rising; only some industries, however, are experiencing this loss in some degree, and not every business is without the buffers and safeguards to prevent the impact. It is clear that the effect is significant, but it is uneven – more workers are leaving STEM jobs, it's empirically documented, and it's happening faster in some jobs than in others; and, while the impact is real, it's not one that every business is without buffers and safeguards to prevent.

Keywords: *artificial intelligence, automation, mid-skill labour, service industries, job polarisation, routine-biased technological change, labour market, displacement, augmentation*

1. INTRODUCTION

A major issue in labour economics which is subject to intense debate is that of technological change and jobs. For each wave of technology that is introduced, the demand for labor changes. For example, agricultural mechanisation, in the wake of the Industrial Revolution, and

computerisation of routine cognitive tasks in the late twentieth century have altered labor demands. Unlike previous waves, which typically included either one new technology or a handful of newer ones, the current wave is characterized by its speed, vastness and its connectivity to non-manufacturing service sectors that are

thought of as harsher and more difficult to automate.

Today, in advanced economies services dominate work, including the financial, retail financial, insurance, telecom, healthcare administration, legal processing, and customer support sectors. The service sector would contribute about 80% of GDP and about 80% of non-farm payroll employment in the U.S. This enormous industry has been characterized by employing people in a relatively robust middle layer of the workforce, ones engaged primarily in middle-level, routine cognitive activities, the "mid-skill" workers as a class. These are the clerks, the call centre workers, the investigator, the insurance worker, the administrative coordinator's, the rules-governed, sequential and codifiable jobs that these workers do.

The primary research question in this paper is: How much has the introduction of AI and automation also decreased the demand of mid-skill labor in the service

sector? There are three strands of evidence to consider when reflecting on this question; macro-economic (and aggregate) labour market trends; sectoral and occupational evidence of lost and transformed jobs; and firm-level and industry case studies of what firms are doing with AI and what impacts are changing in their industries.

The structure of the paper is as follows. In Section 2, we present the concept of routine-biased technological change and job polarisation, which we define and discuss in theoretical terms. In Section 2, we introduce and discuss the concept of routine-biased technological change and job polarisation in theoretical terms. Section 3 examines the available aggregate evidence on the impacts of AI exposure on labour market outcomes. The service sector is explored in Section 4 in greater detail, drawn from examples from banking, retailing, customer service and administration. Case studies of firms are provided in Section 5. In contrast, Section

6 examines the competing factors—augmentation, new job creation, and the apparent boundaries of AI today. Section 7 considers policy implications and section 8 provides a synthesis.

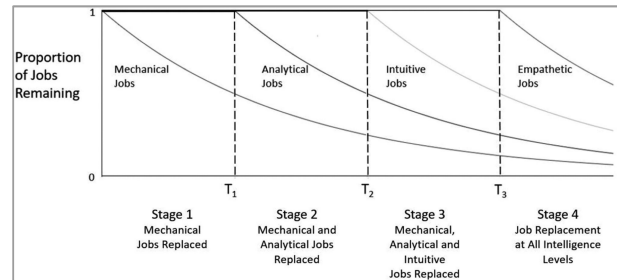
2. THEORETICAL FRAMEWORK: ROUTINE-BIASED

TECHNOLOGICAL CHANGE AND JOB POLARISATION

2.1 The Task-Based Approach

The conceptual support for an understanding on the impact of AI on mid-skilled labour is related to the task-based model of the labour market which was created by Autor et al. (2003) and has been built upon by Acemoglu and Autor (2011). Within this context, work units are broken down into parts or components, not work units as entities or bins of skills. There are two dimensions of tasks: routine, non-routine and cognitive, manual. The main prediction is of course that technology, either ICT or robotics or AI, can replace

most easily the labour used in repetitive tasks and that these tasks "follow explicit rules" that can be programmed into the programs or into robotics or AI.



Huang, M.-H., & Rust, R. T. (2018).

*Artificial intelligence in service. Journal of
Service Research, 21(2), 155–172.*

<https://doi.org/10.1177/1094670517752459>

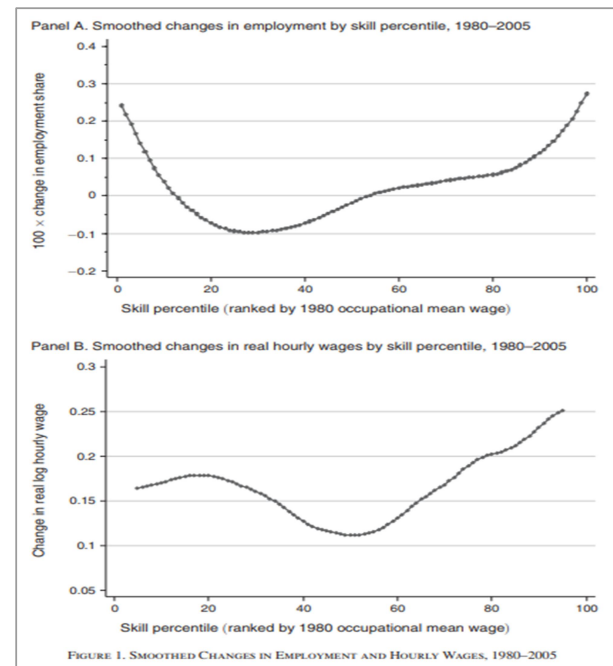
This number represents the way forward in the development of artificial intelligence to replace human labor. AI first automates mechanical tasks, then analytical tasks, followed by intuitive and eventually empathetic tasks. The framework argues that occupations with a higher share of routine cognitive tasks are most at risk in the formative stages of AI: customer service representatives, administrative clerks and bank tellers are among them.

Mid-skill jobs are more likely to be repetitive work. Autor and Dorn (2013, p. 1554) write that "routine task-intensive occupations which include clerical and repetitive production jobs are generally concentrated around the center of the occupational skill distribution, making them the most susceptible to automation. The jobs that are harder to substitute for are those that require low-skill personal service and manual occupations, with tasks involving interpersonal judgement or creativity; high-skill workers, in contrast, have a lower threat of substitution due to tasks of abstract reasoning and creativity, and lack physical dexterity or unstructured environments.

2.2 Job Polarisation: The Empirical Record

The resulting pattern of 'job polarisation', whereby in countries such as Europe and the US, job growth in the higher and lower income bands increased, whereas middle-income job growth stagnated or decreased,

during 1980-2010 has been well established in the empirical literature.



Autor, D. H., & Dorn, D. (2013). The growth of low-skill service jobs and the polarization of the U.S. labor market. American Economic Review, 103(5), 1553-1597. <https://doi.org/10.1257/aer.103.5.1553>

The above graph shows the phenomenon of job polarisation in United States between 1980 and 2005. Panel A displays the employment growth trends across the low-skill, middle-skill, and high-skill categories, with high-skill and low-skill

having had the largest gains in share of employment and the highest percentage growth, respectively. The employment shares grew in both low-skill and high-skill occupations, and even increased in middle-skill occupations, though the percentages were not as high; as seen in Panel A. As can be seen in Panel B, the wage growth picture is similar with wages of middle-wage occupations showing the lowest growth. The figure dovetails with the theory of routine-biased technological change, which predicts that routine cognitive and clerical occupations (usually held by workers of medium skills) would become more likely to be replaced by computers and automation.

Employment in the United States, in the lower middle skill level group, declined significantly from 1980 to 2010. Nearly half of all workers engaged in lower middle skill occupations in 1980, the majority of which were machine operators and administrative support workers. Job share arrangements began to grow during

this time period, particularly among those with high-skills, and shrank for those in the middle skill range, but continued to grow among the high-skill job share holders (Federal Reserve Bank of New York, 2012). The data behind the IMF Working Paper by Bluedorn et al., (2019) further confirmed the existence of life-cycle profiles, which highlighted job polarisation trends, were strongly evident in the UK for low-skill women, and showed that overall, occupations were being downgraded.

The findings of a study by Goos, Manning, and Salomons (2014) ably confirm the cross-national phenomenon of job polarization from 1993 to 2010, and they attribute about half of this reduction in middle-wage jobs to the routine-biased nature of the technological change, and an additional quarter to offshoring. There is widespread agreement in the labour economics literature that this phenomenon of the middle-skilled job hollowing out is

driven mainly by routine-biased technological change.

Most important, this approach is now coming into effect for service sectors. The routine intensive middle of the service sector becomes vulnerable just as manufacturing and clerical occupations in goods-producing industries have been in recent years: the roboticization of cognitive routine work — data processing, form management, answering customer queries, straightforward financial suggestions.

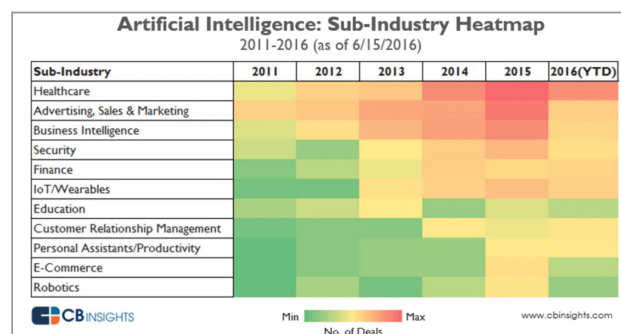
The theory lends further support to the idea that routine cognitive service occupations, including data entry and data processing, call scripting, and simple financial analysis, represent the largest category of service occupations and are most vulnerable to AI substitution. Contrast, however, with non-routine cognitive tasks that require interpersonal judgment, abstract reasoning, or tasks that rely on personal care or trades work, they continue to be far more resistant to

automation. The rest of the analysis looks at the experience of this theoretical vulnerability in the reality of employment results.

3. AGGREGATE EVIDENCE: AI EXPOSURE AND LABOUR MARKET OUTCOMES

3.1 Scale of Potential Displacement

Three institutions have the most recognized and similar estimates of the proportion of jobs that AI could replace: the World Economic Forum, Goldman Sachs, and McKinsey Global Institute.



Faggella, D. (2016). *Artificial intelligence industry: An overview by segment*. Emerj Artificial Intelligence Research.

<https://emerj.com/artificial-intelligence-industry-an-overview-by-segment/>

The image above is a heat map of AI apps by industry segments. The visualization shows continents and countries, with the distribution of AI technologies across service-driven sectors like Finance, Healthcare, Retail, Customer service, and Professional Services in the cross-section of the continent with the industry line. The visualization depicts the spreading of AI technologies: Finances, Healthcare, Retail, Customer service, and Professional services, on the continents and countries using the intersection of the continents with the industries line. This expansive use of AI in repetitive service tasks lends further momentum to the notion that challenges to automation are now ebbing into service and aren't just confined to manufacturing.

The WEF's Future of Jobs Report 2025 says that based on a survey of more than

1000 top global employers that combine over 14 million formal employment across 22 industry clusters of 55 economies, by 2030 job disruption is expected to comprise 22 per cent of the 1.2 billion jobs included in the database. Specifically, it's expected that 170 million new jobs would be created, and 92 million would have been lost in total, leaving the net number of jobs to go up by 78 million. The report reveals that an eye-catching 41 per cent of employers intend to cut back on hiring when other jobs are automated through AI.

Business analysts such as those at Goldman Sachs have been making larger than average estimates of how many full-time equivalent jobs worldwide could be replaced by generative AI, at up to 300 million (Briggs and Kodnani, 2023), and up to 18 per cent of globally spend time on work tasks could reportedly be automated. According to Goldman Sachs, the impact of AI will affect administrative and cognitive tasks in the United States, where

approximately 25% of work hours could be automated.

Importantly, Goldman Sachs pointed out the level of concentration was higher in high-skill professional services, not just in manual jobs — suggesting that disruption is not confined to simple mid-skill jobs, although these are among them.

In the scenario of the midpoint assumption among worldwide companies and accelerated by generative AI, the McKinsey Global Institute project that between 20 and 30 per cent of the working hours currently required will be automatically completed by 2030 all over the world. According to the 2024 AI analysis by McKinsey, ‘AI fluency’ — defined as the ability to use and manage AI solutions — has seen its presence in job postings in the US grow by seven times over the past two years, higher than all other skills, from around 1 million workers in occupations requiring AI fluency in 2023 to around 7 million in 2025. Such a skill premium acceleration is the very fact

of structural labour market adjustment happening.

According to the IMG, just about 40 per cent of all jobs around the world are significantly exposed to AI capabilities, a share that is much higher in advanced, highly digitised economies where service occupations account for a higher share of the economy. Of the total of the current jobs, generative AI is estimated to pose automation level threat to 5.5 per cent of the jobs in high-income countries, compared to 0.4 per cent in low-income countries, reflecting the occupational structure of such countries (ILO, 2023).

3.2 Occupational Evidence: The Mid-Skill Service Worker

In addition to the aggregate impact data, occupational-level data provides a granular data impact on individual and lower level mid-skill service roles. Perhaps instructive, by 2023 the US BLS projects a growth of 464,700 new jobs, of which 390,800 will be nurse roles. Notably, BLS projections

(2023) show that 464,700 jobs will be created, with 390,800 of those to be filled by nurses.

Between 2023 and 2033, digital banking and AI assistants will replace routine transaction work in the role yielding a projected 15 per cent decrease in workforce, with the loss of some 51,400 bank teller jobs. As digital banking and AI virtual assistants handle routine transactions, the number of bank tellers is expected to drop by 15 per cent, losing about 51,400 jobs between 2023 and 2033.

In cashier jobs, with a downtrend of 11 per cent, there will be 353,100 fewer jobs in the same period, as self-checkout and AI-powered systems for paying in retail sectors grow. Customer service representatives are set to experience a 5.0 per cent drop in employment over the 10-year period, medical transcriptionists (4.7 per cent) and credit analysts (3.9 per cent) — all in the same decade.

These projections are in line with those of the OECD Employment Outlook 2023,

which estimated that 27 per cent of employment is at the most strongly threatened level of automation in OECD member countries. The lowest percentages of very high risk jobs are found in economies of Luxembourg, the United Kingdom and Sweden where more of the economy has already adjusted structurally; the higher the percentages of routine intensive jobs in the economy, the higher the level of ongoing structural adjustment in the southern and eastern European economies.

For each occupation, this empirical study leverages payroll dataset from ADP, using task-exposure measures from Eloundou et al., 2023, which quantify the degree of exposure to a specific task. We take advantage of the task-exposure measures from Eloundou et al., 2023, that measure fidelity, or extent of exposure, to a specific task for each occupation and use employment trends from workers in more AI-exposed occupations versus less AI-exposed occupations, as determined from

payroll data from ADP (Brynjolfsson et al., 2025; cited in Anthropic Research, 2026). Entry into the most AI-exposed jobs decreased by about half a percentage point per month in the post-ChatGPT era, with the averaged estimate being a 14 per cent drop in the job-finding rate compared to 2022. One of best causal evidence found so far that connects LLM use to observed labor market outcomes of workers in AI-sensitive jobs.

Based on Site Selection Group data, customer service jobs in the United States have fallen by around 80,000 over the last two years, which spans directly over the considerably accelerated growth and rollout of AI chatbot technology (AIMultiple, 2026). This trend is observed and also it is predicted theoretically, and the occupational forecasts support it to a certain extent.

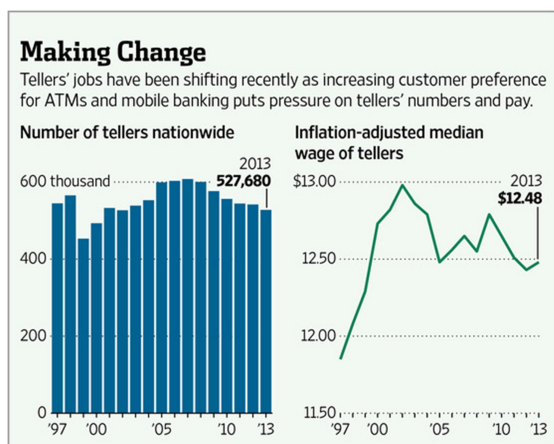
4. SECTORAL EVIDENCE: AI AND MID-SKILL LABOUR IN KEY SERVICE INDUSTRIES

4.1 Financial Services and Banking

One of the best examples of AI taking over part of the middle skill roles comes from the financial services industry. The financial services sector offers some of the most visible examples of AI replacing workers with middle skills. Why algorithmic trading is so important is that it has long displaced stock traders, which were one of the more often mentioned examples, one being Goldman Sachs, which once had 600 stock traders but now, by 2017, occupied the trading floor with a handful of stock traders supported by a cadre of computer engineers working on algorithms. In addition to trading, AI today is reshaping lending and compliance oversight, fraud identification and prevention, loan underwriting, and customer engagement, among other domains — each a domain where

significant mid-skill human engagement was a necessity.

Case studies reported in AI Job Displacement Statistics by SQ Magazine (2026) indicate that financial services processes have successfully reduced manual processes by 30 to 60 per cent, depending on how processes are designed, in light of robotic process automation (RPA) and AI tools. According to the IMF's 2024 study, profits among banks around the globe in advanced economies could increase by 12 to 17 per cent until 2027 due to improved efficiencies brought by AI, costing some \$180 billion due in part to a reduction in the number of people required to perform routine financial activities.



Rexrode, C. (2014, November 17). Bank tellers battle obsolescence. The Wall Street Journal.

<https://www.wsj.com/articles/bank-tellers-battle-obsolescence-1416244137>

The Employment and Wages Trends of Bank Tellers in the USA from 1997 to 2013 are given in above figure. The number of bank Tellers has declined, and this trend has been well documented. This new AI-driven revolution has evolved differently than the ATM introduction of the past, which led to an emergence of the Bessen paradox: as transactions became cheaper, the total number of bank branches grew, including applications for more tellers. Digital banking, mobile apps and AI chatbots are not just about automation of some transactions, but they can lead to physical branches being redundant, reducing the number of jobs for all the people that are working in physical branches. This trend has contributed to the projected 15 per cent reduction in bank

teller jobs by 2033, which translates to 51,400 fewer positions in banking go to bank tellers (BLS, 2023).

4.2 Retail and Customer Service

Retail is another area where there is a high level of mid-skill dislocation. Featuring self-service checkout, sleeker inventory systems via artificial intelligence, and a rise in ecommerce, which diminishes the need for human interactions at the point of sale, the cashier is among the less-skilled services roles where BLS forecasts employment losses – at 11 per cent (353,100 jobs) on an 11-year decline.

The questionable ground is where goes the customer service call centre. Routine questions, such as checking balances and resetting passwords, tracking orders and standard complaints, are being met increasingly by AI-powered chatbots and voice assistants, which typically replace human agents. According to a 2023 survey of US business owners, almost three-quarters of them claimed they've

effectively used AI chatbots to enhance their customer experience, and the sector is estimated to see AI bots assist an additional 40 per cent of fundamental customer questions. Reduced staffing in customer service departments has been cited as the reason for hiring freezes or customer service layoffs in certain companies, which argue that since they are changing to use chatbots for their customer support, they can no longer justify hiring humans in this role. Many companies reported on hiring freezes, or even job reduction in their customer support teams, with the chatbots replacing the human workforce introduced as the reason. But if some said that, it is because they were not able to justify employing people when they now use the chatbots for their customer support teams. BLS anticipates that customer service representative jobs will drop 5 per cent between 2023 and 2033.

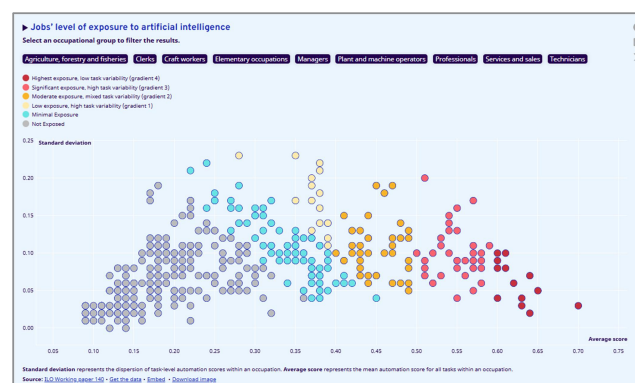
Europe's service sector is much more dynamic than its manufacturing

counterparts are, and in fact an estimated 2.8 million individuals may need to take a different career path by 2030 in order to remain employment in their industry, according to research from McKinsey that defines the need to be found throughout the retail sector more generally. Traditional jobs, like office clerks and executive secretaries, are likely to be first to be displaced by the presence of generative AI, as it is a more powerful alternative to human work productivity. But eCommerce is on the rise and will change how people shop as we know it, prompting a greater need for logistics and digital marketing skills, and alleviating the strain on the workforce in the stores.

4.3 Administrative and Clerical Functions

ILO's Working Paper on Generative AI and Jobs, which focuses on analysis of 436 occupations from internationally standardised ISCO-08 classification, shows that clerical jobs have the highest level of technological exposure among all

major groups of occupations. The majority of clerical jobs were found to have at least some exposure to GenAI automation (over a quarter was highly exposed and over a half medium exposure), with a few jobs identified as having low exposure. (ILO, 2023). It is the most comprehensive cross-national occupational analysis around and its findings are clear on where the pressure from automation is happening the most.



International Labour Organization.

(2025). How might generative AI impact different occupations? *International*

Labour Organization.

<https://www.ilo.org/resource/article/how-might-generative-ai-impact-different-occupations>

The above figure depicts the level of exposure to AI of different occupational categories according to the percentage of specific Artificial Intelligence tasks within their tasks and the level of task variability. Occupations with the highest average AI exposure scores are those that involve routine cognitive and clerical work, while jobs that involve physical work or interpersonal interaction or a range in the kinds of tasks have lower scores. The figure supports this concept being presented as the very argument is: Mid-skill clerical and administrative jobs are most susceptible to AI automation and restructuring.

Like AI, administrative support occupations involve rule-governed, sequential processes that are information processing driven, from insurance, healthcare, government, and legal services to their back-office roles of data-entry clerks, scheduling coordinators, payroll administrators, and document processors. To show the impact its AI tools can have,

here's an example: IBM's own AskHR system uses artificial intelligence to answer 94 per cent of the company's employee HR-related queries immediately without needing a human touch, processing 11.5 Million such queries each year. A cut in IBM HR staff was part of the implementation of this system.

This displacement has a gender aspect and it is important, but not well recognised. According to ILO (2023), over half twice the proportion of female work is at risk compared to men. This is because a high proportion of women is over-represented in clerical occupations, especially in high and middle-income countries. Clerical disruptions have historically been a key driver of women's access to work and AI skill upsets in this area have consequences for gender equality in the labour market.

4.4 Legal and Professional Services

In areas like professional services, where jobs are traditionally high-skill and involve low routine, AI is taking a bite out of the

demand for mid-skill roles. Legal research assistants, paralegal document reviewers, compliance analysts (entry-level), financial analysts (entry-level) all have relatively high amounts of routine cognitive content: Researchers carry out legal research, document reviewers evaluate fair dealings in legal documentation, compliance analysts determine whether the template is statistically compliant with the regulations, and human financial analyst enter data.

According to Bloomberg research in the role of AI in Final Round AI (2025), 53 per cent of the work of market research analysts and 67 per cent of the work of sales representatives can be automated using AI, with managerial roles being at risk for automation between 9 and 21 per cent. This lack of symmetry in the professional services sector, in terms of displacement happening in the above mentioned aspects of professional work, has been seen to follow the same pattern as a tinkering with more repeatable tasks,

throughout the various professional experiences, reducing their relative frequency in higher ranks.

The results of generative AI on professional benchmarks instill this worry: GPT-4-class models are performing at the top 10 per cent of all students on bar exam simulations, and provide the correct response to about 90 per cent of US medical licensing questions. Coding systems used by AI achieved a standardised coding benchmark, SWE-Bench, in 2024, solving 71.7 per cent of the problems compared to 4.4 per cent in 2023 (ALM Corp, 2026). These benchmark results don't apply 1-to-1 to job displacement, but they promote the potential of even greater capabilities for AI, areas that were once thought to be exclusively human.

5. FIRM-LEVEL CASE STUDIES

5.1 Klarna: AI-First Workforce Reduction and Its Limits

The service industry's most talked-about case study of dislodgement by AI mid-skill roles is that of Klarna, a Swedish buy-now-pay-later fintech. The company's tech – primarily OpenAI's – has been used to build an AI chatbot that processes two-thirds of customer support interactions without assistance, in late 2023, Klarna's CEO Sebastian Siemiatkowski announced that the company's AI chatbot is handling the equivalent of 700 full-time customer service staff. Klarna cut its FTE by 24 per cent, down from 5,000 to 3,800, in a year through the use of AI deployment, and a hiring freeze (FX Intelligence, 2024), with a stated aim of reaching 2,000 employees. This improvement allowed the company to turn a profit, almost on par with its breakeven level after losing a number of years.

But, the Klarna example also demonstrates the downsides of an AI-first workforce approach. Customer satisfaction levels were significantly reduced and service quality was not consistent by the end of 2024 and at the beginning of 2025. Klarna changed its tune, starting to rehire, and engage software engineers, designers and marketing people to respond to customer questions (Fast Company, 2026). The CEO used to ignore people but got himself into trouble because there was a focus on cost, not experience. While AI can assist in a massive amount of routine interactions, more complex customer issues, escalation handling, and handling of emotional matters can still be significantly human.

5.2 IBM: Targeted Automation in HR Functions

IBM's take on AskHR's AI system is a more measured and augmentative approach. 11.5 million employee HR interactions every year are managed by AskHR, with 94% of them answered automatically. As AskHR grew, about 200

HR staffers were lopped off by 2024, IBM confirmed. In line with the theory, IBM's larger workforce trend has been to cut on-staff staff in back-office and administrative positions while filling complementary higher-skilled positions in engineering, cloud computing, and AI consulting services.

AI's impact at the business level is a two-way process, as with the shift of a few functions for the majority of mid-skill roles, the company also opened up new opportunities for high-skill functions. At the occupational level, the direction of the pressure is obvious, although unclear at the firm level, with jobs in routine-oriented administrative occupations being threatened, while AI specialist occupations are gaining.

5.3 Broader Corporate AI-First Layoff Trends

The Digital Applied (2026) analysis of such layoffs revealed a trend of companies specifically targeting workforce

rationalisation driven by AI automation, rather than merely restructuring, decelerating hiring cycles due to market factors, or making corrections relative to the introduction of COVID-19. Since mid-2024, the number of redundancies attributed to AI in different sectors—such as fintech, e-commerce, education and enterprise software—has been on the rise quarter by quarter. The result is a competitive trend: As AI-productivity papers are launched by organisations such as Amazon and Klarna (with less employees and higher gains), and they come with a side of increased margins, boards and investors at their competitors are asked to do the same.

The losing jobs in question reflect macroeconomic indicators of job-changing trends, as measured by Goldman Sachs data tracking occupations where technology is used. At the same time, the average unemployment rate two decades after college graduation for tech-exposed occupations increased by nearly 3

percentage points, compared to a same-sized increase for tech-exposed occupations overall and for those in other occupations without a tech connection. Meanwhile, the average unemployment rate ten years after graduation has risen by almost 3 percentage points among tech-exposed workers ages 20–30; this is in line with anecdotal evidence that generative AI is causing hiring slowdowns for young college graduates in jobs with technology impacts.

6. COUNTERVAILING FORCES: AUGMENTATION, NEW JOB CREATION, AND AI'S LIMITS

6.1 The Augmentation Argument

The augmentation hypothesis is another approach to this narrative that outlines the idea that AI may not replace jobs but instead augment them instead. The ILO's 2023 Working Paper on Generative AI and Jobs, the largest comparative occupational

study in the country to date, clearly states that the majority of jobs and industries are not fully susceptible to automation, and are rather more likely to be complemented by the new wave of Generative AI. Among all the impact areas, the most important seems to be the quality of the employment opportunities, specifically greater or lesser work intensity and autonomy, and not quite the loss of employment.

In augmentation, AI handles certain components of a job, freeing up the time spent on other, often more complicated, or interpersonal responsibilities. A customer service agent who handles routine inquiries via a chatbot could be spending more time on myriad complex or “difficult” issues with customers, while a bank analyst using a more mundane data compiler might be spending more time interpreting the data and assisting clients with advice. In this view, AI contributes to the creation of new roles or shifts in the nature of existing jobs by augmenting or augmenting tasks.

ILO estimates of displacement risks are currently slightly higher than in 2023, in part because of lower overall automation scores compared to last year (mean automation score in 2025 is 0.29, compared to 0.30 in 2023), which indicates that as people become more familiar with the types of GenAI tools and more practical examples are provided, earlier estimates of displacement risk may have been too high.

6.2 New Job Creation and Skill Complementarity

One trend shared by multiple institutional reports is the emergence of new job roles alongside the disappearance of others that AI will engender. In terms of job creation, WEF Future of Jobs Report 2025 forecasting a net positive in terms of new jobs created of 170 million with 92 million jobs displaced on an aggregate basis. The newly created jobs are centered on AI development and deployment, such as AI engineers, machine learning specialists and data scientists, as well as AI oversight and

governance, like AI auditors, ethics specialists and compliance managers, and the ‘last mile’ of human interaction that AI cannot emulate, like complex customer relations, emotional support, physical trade work and care provision.

According to the McKinsey Global Institute (2024), demand for ‘AI fluency’ has increased sevenfold in the job market for the US from 2023 to 2025. Individuals with specialised AI-related skills in the same job are earning up to 56 per cent more than those who do not have AI skills (DemandSage, 2026). The premium is truly a reflection of the combination of human judgment and the capacity of AI - and the opportunity for middle skill workers with a learning path versed in AI skills.

But the phrase transition argument has had a critical limitation found by several sources: that the jobs it seeks to transition to aren't in the same places, don't take the same skills, and won't be automatically given to the same workers displaced from

mid-skill service jobs. Those who are most likely to be impacted by AI-driven displacement struggle the most when transitioning to an AI-complementary role – older clerical workers, less digitally literate workers, and those working in lower income countries with less reskilling opportunities. Aggregate employment figures can thus hide very high level disparities in employment distribution at the aggregate worker level.

6.3 The Genuine Limits of Current AI

The displacement narrative is also limited by empirical analysis of the capabilities of AI. The study, by researchers at Anthropic (2026) based on over 17,000 assessments by workers of the performance of AI on actual skills that involve job tasks, reveals that "rising tides" - AI capabilities augmenting and speeding up human activity - prevail over "crashing waves" (full job replacement). The study finds in 2025-Q3 that with an autonomy rate of 70 per cent, frontier AI models are able to successfully complete tasks that human

workers would only be able to do on average over one week.

Although there are many professions expected to be impacted by AI, there are "no signs of slowing labour demand" across OECD countries when looking at the aggregate data on the labour market, the OECD Employment Outlook 2023 says. The disconnect between predicted and actual displacement rates stems from a number of practical challenges, including: the costs and complexity of deploying AI at organisational level; the invasiveness of regulation and compliance mandates on certain industries (such as banking, healthcare, etc); the necessity for humans to review and modify AI-generated results; and that AI isn't yet capable of handling some tasks that require contextual judgement, emotional intelligence, and/or physical engagement.

The OECD also highlights that the degree of exposure of industrial sectors and occupations to AI is not uniformly distributed and that high-skill professional

services providers, in addition to the traditional mid-skill service worker, are highly exposed to the impacts of AI — making it hard to tell one segment of the skill "distribution" or skill "layer" from harm.

7. POLICY IMPLICATIONS

Based on the evidence reviewed in this paper, there are a number of concrete policy implications for the transition that governments, business and the education system can implement to enable a fair transition.

7.1 Reskilling and Upskilling at Scale

Investment in reskilling and upskilling programmes is the most strongly-supported policy response on offer in institutional reports. According to the WEF Future of Jobs Report 2025, 85 per cent of companies will be investing in training. As McKinsey estimates between 75 and 375 million workers across the world are going to have to learn new skills

and change their types of jobs by 2030, it is a challenge which is bigger than the capacity of the existing public and private sector training provision. Skills that are growing the fastest: AI and big data (1st placed skill in WEF Employer Survey 2025), networks and cybersecurity, technological literacy, analytical thinking, resilience and leadership.

Mid-skill service workers face a unique transition problem due to the distinctiveness of what skills are going to be in demand in AI-complementary jobs and what is still going to be valuable in routine-intensive jobs. In fact, a bank teller's knowledge of the mechanics of a daily cash transaction doesn't easily translate into a home for AI model oversight or cyber security. Reskilling programmes need to, therefore, encompass a holistic component of tapering of cognitive and interpersonal skills and cover not only digital literacy training but a more company-wide process which will

take longer and cost more than retraining for neighboring jobs.

7.2 Social Protection and Transition Support

ILO (2023) makes a specific call to governments and employers to ensure that social protection and access to retraining is covered in situations of displacement. This is especially crucial because mid-skill services displacement tends to happen differently for men and women, with women bearing the brunt of being displaced from the majority of clerical and administrative tasks, leading to the need of gender-specific policies such as flexible reskilling pathways, targeted income support and offer of childcare to enable participation in programmes to ensure that AI is not exacerbating existing gender inequalities.

7.3 Regulation for Responsible AI Deployments

However, several authors warn that the prospect of AI leading to the loss of jobs is

not inevitable. The adoption rate and level of adoption in service sectors is additionally influenced by institutional norms, regulation, and firm and employee choices. The pace of substitution is slower in certain areas such as financial advice, legal services, and healthcare, where human oversight and regulatory oversight are more critical in execution and more challenging to replace. The role for human oversight and accountability is important in sectors like financial advice, legal services, and healthcare, as the pace of substitution is slower and a structural buffer to rapid displacement. The governance of AI – such as human oversight over costly or important decisions influenced by AI, transparency in employment practices driven by AI, and workers' participation in decisions making regarding AI deployment – can profoundly impact distributional impacts without compromising productivity benefits.

7.4 Investing in Sectors AI Cannot Easily Replicate

The ILO (2023) further calls for more investments going into lesser-funded sectors that can serve as sources of 'good quality jobs' that are less vulnerable to automation including the care sector, education, and physical trades. These are the non-routine, interpersonal and physical-dexterity tasks, which as yet AI can deal with poorly and also are areas with high demand in ageing societies. Investment in these industries strategically, such as building buildings, early childhood education centres and schools, can provide opportunities for employment for those working in more mundane service positions and help to fulfill social needs that are not met through market forces.

8. SYNTHESIS AND OVERALL ASSESSMENT

In relation to its main research question: How much has the introduction of AI and

automation impacted on the demand for mid-skill workers in service industries? the evidence presented in this paper quantifies the impact as significant and increasing, but has, in several respects, a more nuanced, partial and task-specific character that has gone unappreciated by the more doomstoned narratives.

Autor, Levy, Murnane, Acemoglu, and colleagues have developed a theoretical framework that they have called routine-biased technological change; it has proved to be a long-standing and empirically valid lens. People in service occupations, particularly routines involving cognitive tasks, like in customer support, banking, retail and administrative and back-office professional roles, are the sweet spot of the job distribution most likely to be replaced by AI. Indeed, the empirical record of the 1980s to 2010s shows sustained job polarisation precisely in line with this prediction, and the current wave of use of the AI is both increasing the rate at which

jobs polarise and pushing them further in that direction.

At sectoral and occupational dimensions there is more tangible evidence that by 2030 jobs at these levels will decline by 15 per cent in banks, 11 per cent in cashiers and 5 per cent in customer service representatives, while more occupations will be broad subject to automation pressure in the field of clerical and administrative roles (ILO, 2023; BLS, 2023). Between 2022 and 2024, the United States had a decrease in customer service jobs of 80,000. The job-finding rates for the most AI-exposed occupations declined as estimated by an average of 14 per cent after ChatGPT (Goldman Sachs, 2026) and tech-exposed youth unemployment has risen (Brynjolfsson et al., 2025).

Firm level case studies prove that the power of AI extends to the workforce, most notably in Klarna's 24 per cent sweep of customer service headcount due to the automation of these areas by AI. The Klarna case also shows that the current

power of AI – with its ability to deal with complex interactions, service quality, and replacing knowledge held by humans – is real and significant, and that pushing too hard on an AI-only workforce policy could lead to deterioration in the organization.

Both ILO and OECD recommend that, at the present time, augmentation outweighs the total amount of jobs being replaced, while WEF uses its final message to say that the overall outlook of non-recorded jobs is positive. According to WEF, by 2030, there are 78 million jobs that will be created worldwide, but ILO reaches a different conclusion that most jobs exposed to automation are in the partial, not total, category. These overall evaluations are right, but they tell only part of the story: the new jobs and role types are not the same as the jobs and role types that are losing. They can be located in different organization, employer and physical locations, and the employers may require different skills than those that are losing, generating substantial transition

costs for the people that are losing the jobs, costs not captured by aggregate data.

The reduction of the need for mid-skill labour in service occupations, then, should be seen as: meaningfully and measurably reduced; routine-cognitive task based; predicted to increase as AI gets better; unevenly distributed by gender, geographical region, age, and educational level; and highly influenced by policy choices, regulation, and organizational strategy choices that are within human control. AI and automation have not replaced mid-skill service, they've simply moved it out of the spotlight by decreasing its demand relatively over time and will decrease its demand further, on a firmer basis.

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